GabayGamot: A Web-Based Medicine Locator and Donation Management System for Improving Medicine Accessibility in the Philippines

CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

1.1 Background of the Study

Access to essential medicines remains one of the most persistent barriers to adequate healthcare in the Philippines. Households continue to carry the largest share of health spending. Out-of-pocket payments made up 44.7% of current health expenditure in 2022 (Philippine Statistics Authority [PSA], 2023) and 44.4% in 2023 (Palencia, 2025), and medicines account for roughly half of household out-of-pocket health spending (Clarete & Llanto, 2017). The burden is heavier for the poor: in 2012, the poorest households spent about 76% of their out-of-pocket health spending on drugs and medicines (Ulep & Cruz, 2013, as cited in Magno, 2022). Affordability remains a problem even after price regulation. A 2010 survey of 600 respondents in Caloocan, Manila, and Quezon City found that only 15 could afford the medicines in their prescribed treatment packs (Clarete & Llanto, 2017). A more recent study found that some maintenance regimens for hypertension still cost more than a day's wage (Lat et al., 2025).

Beyond cost, medicines are often unavailable where patients first seek care. Only about two in every ten Philippine hospitals have the complete set of 16 essential medicines, and when a medicine is unavailable, patients must buy it outside the facility at their own expense (Philippine Institute for Development Studies [PIDS], 2022). In one study, essential antihypertensives were available in only 12.96% of public primary-care drug outlets, compared with 60.32% of private outlets (Lat et al., 2025). These gaps suggest that even individuals who can afford their medicines are often left to search across several pharmacies for stock,

particularly for maintenance drugs and specialized medications. Tinam-isan et al. (2024) also identified the lack of a centralized system for checking whether prescribed medicines are available at different pharmacies as a gap in the Philippines.

Several organizations and government programs have stepped in to bridge this gap. Among non-profit organizations, the Philippine Cancer Society, Inc., a private, non-stock, non-profit corporation (Philippine Cancer Society, n.d.-b), accepts requests for medicine assistance from cancer patients. Applicants must submit a prescription and a barangay certificate of indigency, among other documents (Philippine Cancer Society, n.d.-a). On the government side, assistance is offered by the Philippine Charity Sweepstakes Office (PCSO), a government-owned and controlled corporation chartered under Republic Act No. 1169 (1954), through its Medical Access Program (Pineda, 2024); by the Department of Social Welfare and Development (DSWD), through its Assistance to Individuals in Crisis Situation (AICS) program (Department of Social Welfare and Development Field Office I, n.d.); and by the Department of Health (DOH). Republic Act No. 11463 also established Malasakit Centers in DOH hospitals and the Philippine General Hospital as government one-stop shops for medical and financial assistance from the DOH, DSWD, PCSO, and PhilHealth (Malasakit Centers Act of 2019, 2019).

Newer programs have widened this support. PhilHealth's Guaranteed and Accessible Medications for Outpatient Treatment (GAMOT) program provides up to ₱20,000 a year in outpatient medicines to members registered under its YAKAP primary-care program (Philippine Health Insurance Corporation [PhilHealth], 2026). The DOH's Medical Assistance for Indigent and Financially Incapacitated Patients (MAIFIP) program is set to cover FDA-approved medicines under its 2026 guidelines (Philippine News Agency, 2026).

However, these efforts largely operate independently of one another. Philippine health financing remains fragmented, with medical assistance programs often operating in isolation from PhilHealth and from national and local budgets (Pantig & Ulep, 2025). No shared or centralized system connects donors, pharmacies, partner organizations, and the beneficiaries who need assistance the most. As a result, medicine donations are not always matched efficiently with the individuals who need them, and beneficiaries often have no single, reliable platform to search for available assistance.

This lack of coordination and centralized information results in inefficiencies on both ends of the system. Pharmacies and donors with available medicine supplies have no streamlined way of reaching the people who need them, while financially disadvantaged individuals are left to navigate a fragmented landscape of assistance programs, often through word-of-mouth or manual inquiry.

To address this gap, the researchers propose GabayGamot, a full-stack web-based Medicine Locator and Donation Management System designed to connect pharmacies, donors, partner organizations, and beneficiaries under a single, centralized digital platform. Partner organizations include non-profit organizations and government assistance programs. The system aims to make essential medicines more accessible by allowing users to locate available medicines across participating pharmacies using up-to-date stock information. It also provides a structured and transparent process for managing medicine donations from collection to distribution. GabayGamot is intended to complement existing government programs rather than replace them, by helping beneficiaries find which channels of assistance are available to them.

This study focuses its implementation and testing within Metro Manila (National Capital Region), where participating pharmacies, partner organizations, and prospective beneficiaries will be engaged for system evaluation.

1.2 Statement of the Problem

General Problem

How can a web-based Medicine Locator and Donation Management System improve access to essential medicines in the Philippines?

Specific Problems

1. How can financially disadvantaged individuals gain better access to essential medicines?
2. How can users easily identify pharmacies where their required medicines are available?
3. How can medicine donations from pharmacies, donors, and partner organizations be managed through a centralized platform?
4. How do target users and IT experts rate the system in terms of functional suitability, interaction capability, and reliability based on ISO/IEC 25010:2023?

1.3 Objectives of the Study

General Objective

To design, develop, and evaluate GabayGamot, a full-stack web-based Medicine Locator and Donation Management System that improves access to essential medicines among Filipinos, particularly financially disadvantaged individuals, by connecting pharmacies, donors, partner organizations, and beneficiaries.

Specific Objectives

1. To develop a Beneficiary Assistance Module that allows financially disadvantaged individuals to register, submit medicine requests with supporting documents (e.g., medical prescription, certificate of indigency), and get matched with relevant assistance programs (e.g., Philippine Cancer

Society, PCSO Medical Access Program, DSWD AICS, Malasakit Centers).

2. To develop a Medicine Locator Module that allows participating pharmacies to update their stock levels, and allows users to search for medicines by brand or generic name, view participating pharmacies with up-to-date stock information, and filter results by location and price range.
3. To develop a Donation Management Module that allows donors (individuals, pharmacies, and organizations) to list medicine donations with expiry date and quantity validation; allows administrators to verify, approve, and route donations to matched beneficiaries; sends automated alerts for near-expiry and low-stock medicines; and provides a dashboard with reports on donation flow, request fulfillment rates, beneficiaries served, and medicines distributed.
4. To evaluate the system's software quality in terms of functional suitability, interaction capability, and reliability based on ISO/IEC 25010:2023 (International Organization for Standardization [ISO] & International Electrotechnical Commission [IEC], 2023), through testing with target users, including beneficiaries, pharmacy staff, and partner organization coordinators, and with IT experts.

1.4 Significance of the Study

This study is beneficial to the following:

- **Beneficiaries / Indigent Patients.** The system provides a direct and centralized channel through which financially disadvantaged individuals can request and receive assistance for essential medicines.
- **Pharmacies.** Participating pharmacies gain visibility to a wider network of users searching for medicine availability, and a structured channel for contributing to donation efforts.

- **Donors and Non-Profit Organizations.** Individual donors and non-profit organizations such as the Philippine Cancer Society, Inc. are given a shared platform to coordinate and track donations more efficiently.
- **Government Agencies and Programs.** The DOH, the PCSO, Malasakit Centers, and local government units can use the system as a supplementary tool for referring beneficiaries and for monitoring medicine accessibility and donation distribution at the community level.
- **Future Researchers.** This study may serve as a reference for future works related to healthcare information systems, donation management platforms, and medicine accessibility solutions in the Philippines.

1.5 Scope and Limitations of the Study

This study covers the design, development, and evaluation of GabayGamot, a web-based system composed of three modules: the Beneficiary Assistance Module, the Medicine Locator Module, and the Donation Management Module. The system will initially be piloted with selected pharmacies, partner organizations, and prospective beneficiaries within Metro Manila.

The study is limited to managing information on medicine availability and donation coordination. It does not include the actual physical delivery of medicines, courier logistics, or payment-gateway processing for medicine purchases. The system does not replace licensed medical consultation, diagnosis, or prescription issuance, and all medicine requests remain subject to verification and approval by administrators and partner organizations. The accuracy of medicine stock and donation data also depends on how promptly participating pharmacies and donors update it. The stock information shown therefore reflects each pharmacy's latest update rather than live inventory.

Contributions / Individual Task:

Reyes - Project Paper

Sta. Rosa - Low Fidelity Prototype

Tabuzo - Project Paper

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